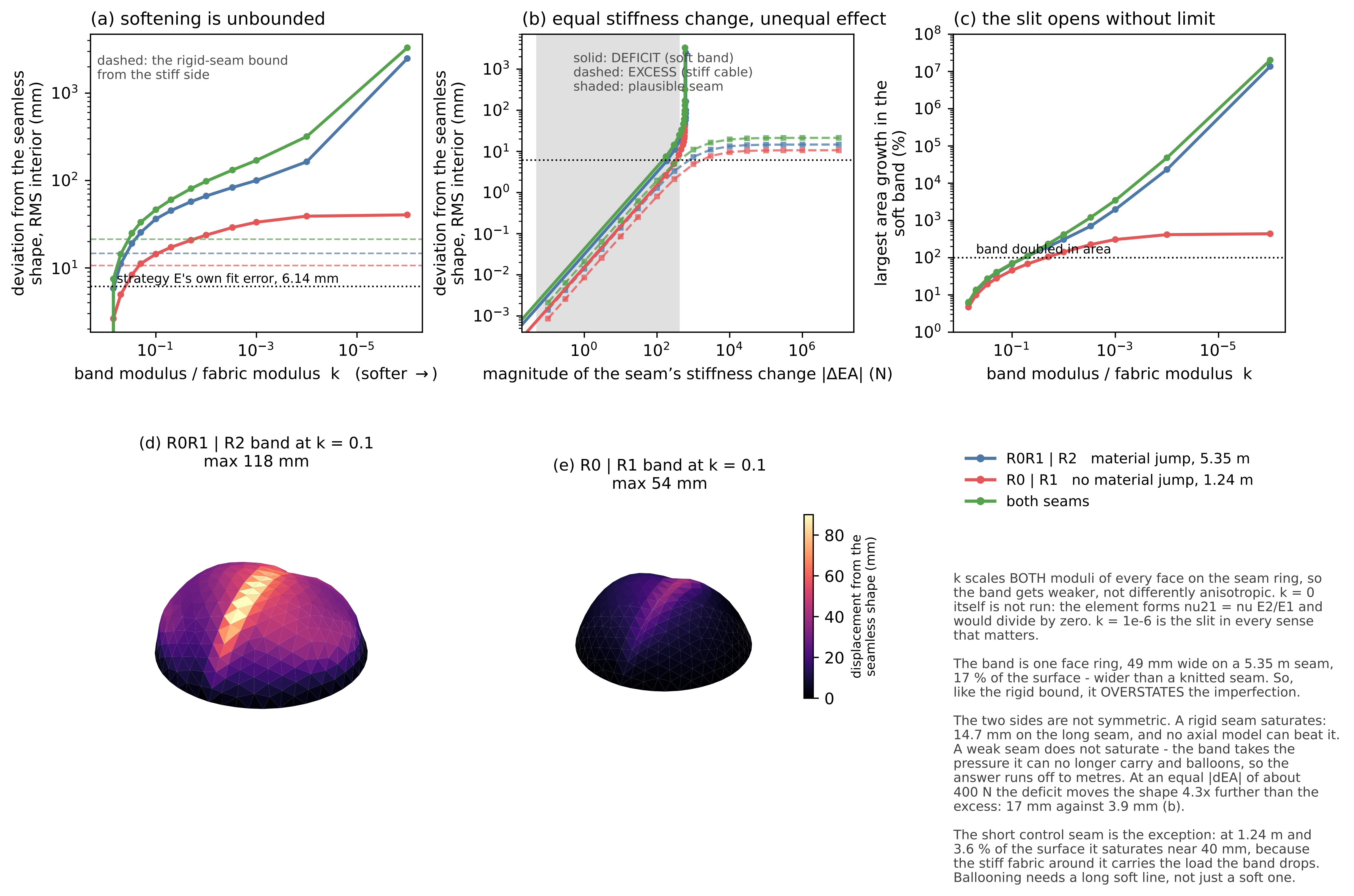
(a) softening is unbounded
dashed: the rigid-seam bound
from the stiff side
strategy E's own fit error, 6.14 mm
deviation from the seamless shape, RMS interior (mm)
band modulus / fabric modulus k (softer →)
(b) equal stiffness change, unequal effect
solid: DEFICIT (soft band)
dashed: EXCESS (stiff cable)
shaded: plausible seam
deviation from the seamless shape, RMS interior (mm)
magnitude of the seam's stiffness change |ΔEA| (N)
(c) the slit opens without limit
band doubled in area
largest area growth in the soft band (%)
band modulus / fabric modulus k
(d) R0R1 | R2 band at k = 0.1
max 118 mm
(e) R0 | R1 band at k = 0.1
max 54 mm
displacement from the seamless shape (mm)
R0R1 | R2 material jump, 5.35 m
R0 | R1 no material jump, 1.24 m
both seams
k scales BOTH moduli of every face on the seam ring, so the band gets weaker, not differently anisotropic. k = 0 itself is not run: the element forms nu21 = nu E2/E1 and would divide by zero. k = 1e-6 is the slit in every sense that matters.
The band is one face ring, 49 mm wide on a 5.35 m seam, 17 % of the surface - wider than a knitted seam. So, like the rigid bound, it OVERSTATES the imperfection.
The two sides are not symmetric. A rigid seam saturates: 14.7 mm on the long seam, and no axial model can beat it. A weak seam does not saturate - the band takes the pressure it can no longer carry and balloons, so the answer runs off to metres. At an equal |dEA| of about 400 N the deficit moves the shape 4.3x further than the excess: 17 mm against 3.9 mm (b).
The short control seam is the exception: at 1.24 m and 3.6 % of the surface it saturates near 40 mm, because the stiff fabric around it carries the load the band drops. Ballooning needs a long soft line, not just a soft one.